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Practical Application of Digital Twins for Transportation Asset Data Management: Case Example of a Safety Hardware Asset

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Abstract

Digital Twins have emerged as a transformative technology with immense potential for enhancing asset data management in various industries. The transportation industry relies heavily on properly functioning and maintaining safety hardware assets, such as crash barriers, guiderails, and road signs, among others, to ensure the safety of road users. However, traditional asset data management practices for ancillary assets often fall short of providing real-time, comprehensive insights into the condition and performance of these critical assets. Conversely, Digital Twins can analyze data for predictive and preventive maintenance using cutting-edge technologies. This will enable asset managers to proactively address potential safety concerns, plan maintenance schedules efficiently, and optimize resource allocation. This can be achieved by leveraging historical asset data and real-time condition updates and providing the Digital Twin with an information-rich analytical model. The key steps to creating an information-rich analytical model are identifying the asset data that should be collected, understanding data architecture throughout the asset lifecycle, and generating asset data governance and stewardship frameworks. As such, this paper investigates the current state of practice for ancillary asset data collection and management and investigates the generation of information-rich analytical models for guiderails, a critical roadway safety hardware. Moreover, this study demonstrates how Digital Twins can be integrated and affect the development of transportation asset management systems. The findings of this article emphasize the significance of adopting Digital Twins in transportation asset data management. The case example showcases a compelling roadmap for other transportation agencies to consider integrating digital twins into their asset management strategies.

Keywords

infrastructure, infrastructure management and system preservation, maintenance management systems, asset inventory, data collection, maintenance data modeling, tactical asset management, maintenance asset inventory, maintenance data collection, infrastructure condition assessment, maintenance work planning

In collaboration with the public and private sectors, the United States Department of Transportation (USDOT) is developing a roadmap for deploying intelligent transportation systems (ITS) (*1*). Such a roadmap aims to transform transportation systems into more digitized and intelligent networks to support safer construction and operation, allow for cooperative automated transportation and faster freight deliveries, improve mobility equity and operational efficiency for an overall better customer experience, and facilitate the inclusion of connected and autonomous vehicles (*1*). One of the primary goals of ITS is to achieve zero fatalities on roads, making

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roadway safety highly important. Tragically, traffic crashes have claimed the lives of over 370,000 individuals in the last decade, accounting for 94.2% of total transportation-related deaths in the U.S.A. Also, the COVID-19 pandemic has coincided with a concerning increase in traffic fatalities, prompting the USDOT Secretary to announce the National Roadway Safety Strategy (NRSS) in January 2022 (*2*). The NRSS is a comprehensive plan that leverages funding provided by the Infrastructure and Investment Job Act (IIJA) and aims to achieve zero fatalities on the nation's roads in the long term (*2*).

A safer road environment can be achieved by digitizing the transportation system and implementing innovative technologies (*3*), where digitization-related technologies play a critical role in the digitalization of the transportation sector (*4*). Digitization can be defined as the full digital creation of information without needing a physical or analog counterpart. Conversely, digitalization can be understood in the context of developing processes and changing workflows and can be defined in two folds: (1) the utilization and implementation of digital technologies within various individual, organizational, or societal contexts and (2) the effects on individuals, organizations, or society as a whole, resulting from this utilization (*5*).Implementing information technologies such as building information modeling (BIM) and geographic information systems (GISs) in integration with the concept of Digital Twins are fundamental for managing transportation assets (*6*, *7*); namely, Digital Twins is promising in supporting asset data digitization and the digitalization of asset management practices and workflows to optimize the performance and maintenance of transportation asset systems, including pavements, bridges, tunnels, railways, and a wide variety of ancillary asset classes (*8–10*).

Traditionally, state departments of transportation (DOTs) have focused on maintaining visible and high-value transportation assets, such as bridges and pavements (*11*). However, it is essential to recognize that transportation systems encompass a range of ancillary asset classes critical to highway safety and operations, such as communication signals, signs, guiderails, and pavement markings, among others (*10*, *12*). Incorporating information technologies and Digital Twins in the transportation sector can greatly enhance the deployment of the ITS future roadmap and achieve zero fatalities. This implementation will enable transportation authorities to gain better insights, optimize safety measures, and move closer to realizing a safer and more efficient transportation networkD. As such, this research paper is part of an overarching objective of investigating innovative solutions by integrating different concepts and technologies, including Digital Twins, BIM, GIS, the internet of things (IoT), computer vision, and machine learning methods to support state DOTs in achieving the intelligent management of transportation assets and conducting complex analysis to support decision-making. Also, this research aims to highlight cross-asset and historically ignored systems, namely the ancillary asset systems, which should be investigated to optimize the benefits of Digital Twins in attaining a holistic management approach and achieving the vision of ITS. This ongoing effort investigated the management of 34 ancillary asset classes; however, this paper closely examines the practical implementation of Digital Twins for roadway safety hardware and its impact on developing roadway safety hardware management systems. It aims to identify various asset data sources, data collection practices, supporting technologies, understand asset data governance and stewardship frameworks throughout the asset lifecycle, recognize the associated challenges, and provide recommendations and future research opportunities.

Background

Roadway Safety Hardware: Overview and Challenges

Roadway safety hardware is considered the last line of defense against roadway incidents and fatalities (*13*). Roadway safety hardware includes longitudinal barriers (such as guiderails), barrier terminals (such as guiderail end treatments), crash cushions, and breakaway hardware (such as signs, signals, and light posts) (*13*).

The Federal Highway Administration (FHWA), in collaboration with the American Association of State Highway Transportation Officials (AASHTO), is continuously working to improve roadside safety hardware. They are considering a three-fold approach: (a) establishing rigorous and standardized crash testing standards, such as the AASHTO's Manual for Assessing Safety Hardware (MASH) to guide manufacturers in developing safety hardware and take into account the wide range of vehicles and consider potential crashes with a diverse range of conditions (*14*); (b) collaborate with local agencies and state DOTs to investigate the proper device installment and maintenance practices of roadside safety hardware and offer support through specialized training programs (*13*); and (c) develop in-service performance evaluation (ISPE) programs to monitor how these devices are performing continuously, collect safety hardware-related crash data and communicate with state agencies with respect to crash reporting, and manage data to optimize the effectiveness of hardware inventory and maintenance practices (*13*).

Although these practices and safety approaches are continuously updated, the unsettling aspect of safety hardware fatalities lies in the heightened crash severity they entail. The main objective of roadway safety hardware is to reduce severe crashes by judiciously implementing barriers and ensuring their effectiveness when

shielding roadside obstacles or terrain features; however, in 2021, approximately 20% of motor vehicle crash fatalities occurred when a vehicle veered off the road and collided with a stationary object alongside the roadway (*15*). During such incidents, the most frequently struck objects are trees, utility poles, and traffic barriers, where 18% of the fatalities happened on interstate and freeways, 38% on other major roads, and 42% on minor roads (*15*).

A report by the U.S. Government Accountability Office (GAO) reviewed the FHWA's framework for roadside safety hardware. It investigated the extent to which information is available on roadside safety hardware's performance once installed and found that ISPE information is limited. On the other hand, having strong and complete data sets is crucial for ISPE and asset management analysis. This allows researchers to thoroughly examine the safety equipment and evaluate its performance in different roadside and crash scenarios (*16*). With this valuable information, researchers can expedite the ISPE process and gain more accurate insights into safety equipment installation and maintenance needs (*16*). As such, FHWA advises state DOTs to improve their asset management practices and consistently track the safety hardware on roadways by gathering and maintaining high-quality information for data-driven decision-making. In addition, FHWA encourages the inclusion of safety assets in transportation asset management plans (TAMPs) to acquire quality information and provide data analysis for an optimized ISPE program (*16*).

State DOTs are facing several challenges in managing safety hardware effectively, where the current practices for roadway asset management primarily focus on identifying asset locations through periodic inventory schedules (e.g., annual or every five years) (*10*, *16*). Collecting this data is challenging, leading state DOTs to seek help from service providers equipped with mobile technologies such as LiDAR and navigational sensors; however, these practices often overlook roadside hardware assets (*16*). Roadside safety hardware data, such as installation dates, installers, and repair history, is not effectively collected or recorded by state DOTs and service providers (*10*, *16*). Having this information could enhance integration into safety asset management programs. However, maintaining an up-to-date database poses difficulty since the accuracy and availability of such data sets are highly dependent on scattered hardcopy documentation across various departments and may require additional information requests from manufacturers and installers (*16*). Also, the lack of data sharing hinders the comprehensive management of such assets (*16*). Implementing technologies for transportation asset management and improving state DOTs' digital capabilities have proven beneficial (*17–19*). As such, it is critical to investigate the implementation of emerging technologies, namely Digital Twins, that would support collecting, processing, and analyzing the asset data and provide data accessibility and integration with management systems.

Digital Twins for Transportation Asset Data Management

State DOTs adopt data-intensive asset lifecycle management systems to support decisions that seek to (a) effectively operate and maintain physical assets, (b) enhance and expand their service life, and (c) optimize their performance (*11*). Thus, with the Digital Twins' capabilities of connecting the enterprise asset data and extracting actionable insights (*10*, *20*), implementing Digital Twins for transportation asset management will support state DOTs in leveraging their transportation asset management practices and improving their asset management maturity.

Ammar et al. (*9*) defined Digital Twins for infrastructure assets as the digital representation of a physical asset or systems of assets connected bi-directionally with their physical counterparts. In their definition, they emphasized that it is crucial to note that data in the infrastructure sector is incredibly diverse and can be sourced from various locations. These sources include semantic asset data from physical assets and their digital representations, spatial data from the surrounding environment, and data collected by intelligent transportation technologies such as fixed sensors and mobile or remote transportation data collection technologies (*9*). The authors further emphasized that Digital Twins can integrate historical asset data and updated asset condition information to create data-rich analytical models, which can then be used with technologies such as artificial intelligence and machine learning to facilitate predictive and preventive maintenance, thus supporting the operation and management of infrastructure assets (*9*).

Implementing Digital Twins for transportation assets has mainly focused on managing pavement performance and predicting the performance of highway materials (*21–24*), supporting a lifecycle management analysis for tunnels (*25*, *26*), developing a Digital Twin-driven management system to support bridge maintenance (*27*, *28*), and deploying Digital Twins as a traffic management and control system for railways (*29*, *30*). More recently, the authors of Ammar et al. (*9*, *10*) investigated the current state of practices of state DOTs toward digitizing ancillary transportation asset data to implement the vision of Digital Twins to support transportation asset management. Also, they contextualized the state DOTs' maturity in the advancement of digital processes.

Integrating Digital Twins in the transportation sector can enhance the effectiveness of managing ITS

components, foster the creation and utilization of digital technologies and programs, and facilitate the optimized digital management of transportation projects from start to finish (*4*). Moreover, Digital Twins can leverage the value of asset information and, thus, improve the process of data-driven asset management decisions. However, the impact of Digital Twins on the development of asset management programs within transportation organizations still needs to be addressed (*31*).

Research Gaps, Objectives, and Significance

Previous research investigated the benefits of implementing Digital Twins for transportation asset management. Still, the impact of its implementation on the development of transportation asset management systems (AMSs) has not been investigated. On the other hand, state DOTs are facing challenges in managing the data of their safety hardware assets, and the data necessary to conduct ISPE is limited. To contribute to filling this gap, this study is part of ongoing research to investigate and document the state of practice in collecting and managing ancillary transportation asset data to investigate the readiness of state DOTs to adopt information and communication technologies, namely Digital Twins, and to support and guide state DOTs in navigating this journey. Since the ancillary asset systems include various asset classes that cannot be comprehensively investigated within one study, this paper aims to highlight a roadway hardware asset where the guiderail was selected as the asset of interest. As such, this study aims to achieve the following objectives: (1) assess the current practices of state DOTs in collecting and managing the data for guiderails, a roadway safety hardware asset; (2) understand and document where state DOTs stand in collecting and digitizing the data for guiderails; and (3) investigate the impact of implementing the vision of Digital Twins in managing the data of a roadway hardware asset on the development of AMSs.

It is crucial to comprehend and record how state DOTs collect and manage data related to roadway safety hardware assets. This understanding is vital for enhancing roadway safety, pinpointing data deficiencies essential for informed decision-making, ensuring compliance with FHWA guidelines, and establishing a centralized knowledge hub for state DOTs to handle the data for roadway safety hardware assets effectively. This will enable state DOTs to proactively approach roadway safety and their goal of zero fatalities, prioritize their safety investments, continually improve infrastructure and systems to prevent accidents, protect road users, and effectively allocate resources. In addition, investigating the impact of implementing Digital Twins on managing a roadway safety hardware asset can lead to more advanced asset management practices and improve asset management maturity, optimized maintenance strategies, enhanced safety, and better-informed decision-making. Implementing Digital Twins will empower transportation agencies to make proactive and data-driven decisions, ultimately contributing to improved roadway safety and infrastructure efficiency.

Methods

This study is part of an ongoing effort to capture the current practices of state DOTs in collecting and managing their ancillary asset data. As such, a web-based survey was developed, piloted with subject matter experts from the Kentucky Transportation Cabinet (KYTC) and Utah State DOT (UDOT), and approved by the Office of Research Integrity at the University of Kentucky (Institutional Review Board [IRB] # 69802) before distribution. The survey was distributed to the AASHTO Committee on Maintenance members and the FHWA asset management contact list. The survey captured the current practices of state DOTs in managing 34 ancillary asset classes. A total of 45 complete responses were received from 30 state DOTs, as presented in Figure 1. Responses were collected from several divisions within the surveyed state DOT personnel with knowledge of ancillary asset maintenance and operation (59% of the responses), ancillary asset data management and analysis (23% of the responses), and other expertise (18% of the responses). Examples of respondent positions include bureau chiefs, managers, directors, and state-wide engineers.

The operation and maintenance of transportation ancillary assets vary by asset class; accordingly, the needed asset data might vary. For instance, the data needed to manage culverts might vary from those needed for traffic signals management. This paper emphasizes the guiderail ancillary asset class, a roadway safety hardware asset, to establish a deeper understanding of the current practices of state DOTs in collecting and managing the data of roadway safety hardware assets. In addition, this paper aims to thoroughly examine how implementing Digital Twins can affect the development of AMSs for roadway safety hardware. To understand the impact of implementing Digital Twins to manage the data of guiderails, it is critical to develop a deeper comprehension of the data requirements and flow and capture the guiderail asset data lifecycle, data governance, and stewardship frameworks. As such, a case study with subject matter experts from Connecticut DOT (CTDOT) was conducted. A summary of the research methodology is presented in Figure 2.

Web-Based Survey

The conducted web-based survey was distributed over two sections. The first section covered the current practices of state DOTs in collecting ancillary asset data, and

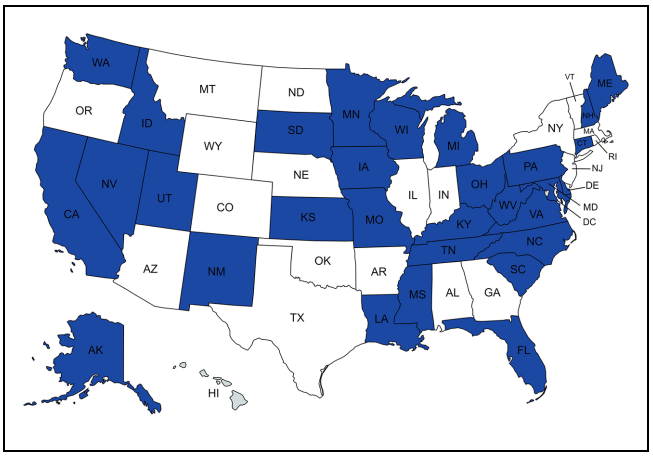


Figure 1. Complete responses received from state departments of transportation (created with MapChart.net).

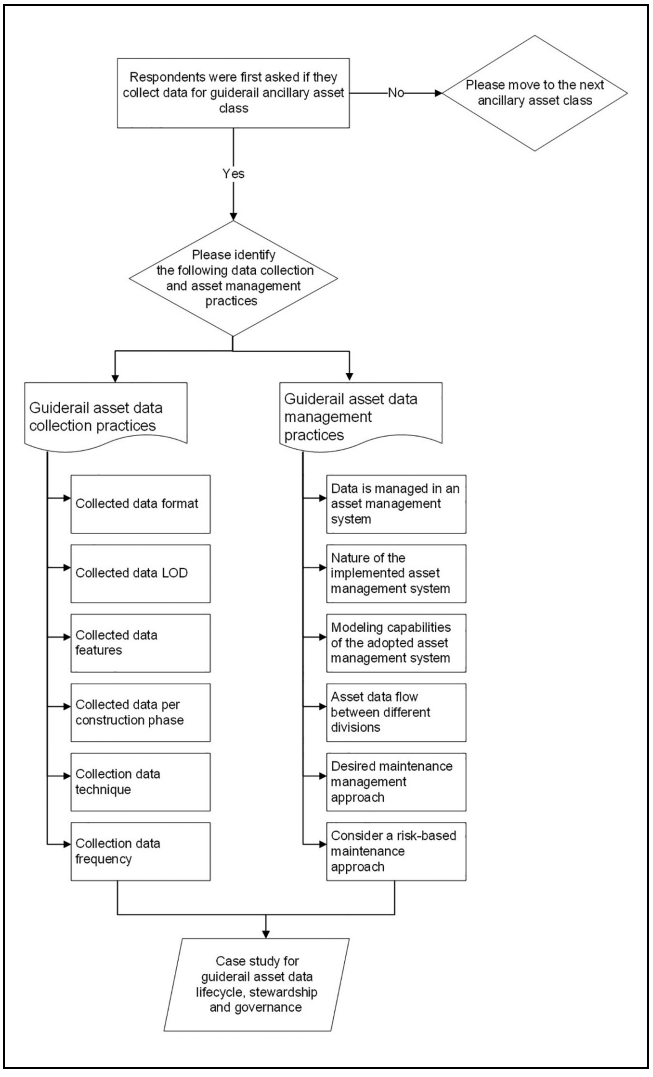


Figure 2. Adopted research methodology.
Note: LOD = level of details.

the second was related to ancillary asset data management. For ancillary asset data collection, six major data collection practices were categorized (i.e., variables), and each category was described by several subcategories (i.e., variable levels). For each ancillary asset, for each variable, respondents filled in a level defining whether and to what extent they collected data related to the variable. Respondents were allowed to select more than one level for some variable levels.

The six major ancillary asset data collection variables and their associated variable levels are as follows.

1. Data format: the format in which the asset data is collected (paper-based, portable document format [PDF], smart PDF [i.e., fillable PDF], two-dimensional [2D] models, and three-dimensional [3D] models).
2. Data level of detail: the level of detail in which the asset data is collected (not available, L4, L3, L2, and L1).
3. Data collection features: the different asset features collected (asset location [i.e., indexed location of the asset], asset dimensions, asset material type and properties, and asset condition).
4. Data collection/construction phase: the construction phase during which the asset data is collected (design, construction, project close-out [as-built], and maintenance).
5. Data collection techniques: the technique used for the asset data collection (manual collection, automated collection, and remote collection).
6. Data collection frequency: the frequency with which the asset data is collected (monthly, quarterly, semi-annually, yearly, more than a year).

The different ancillary asset data collection practices related to variables and variable levels are defined and summarized in Table 1.

Similarly, for ancillary asset data management, six major data management practices were categorized (i.e., variables), and each category was described by several subcategories (i.e., variable levels). For each ancillary asset, for each variable, respondents filled in a level defining whether and to what extent they manage data related to the variable. Respondents were allowed to select more than one level for some variable levels.

The six major ancillary asset data management variables and their associated variable levels are as follows.

1. Asset data is managed using an AMS, to capture if state DOTs use a system to manage ancillary asset data (the agency adopted an AMS to manage the asset data, the agency did not implement an AMS for managing the asset data, the agency did not

Table 1. Ancillary Asset Data Collection-Related Variables and Variable Levels

Variable	Variable level	Variable level's definition	Reference if applicable
Data format[a]	Paper-based; PDF; Smart PDF (i.e., fillable PDF); 2D models; 3D models	na	na
Data level of detail	Not available; L4; L3; L2; L1	***Not available*** (i.e., no identified level of detail) ***L4*** (i.e., basic details with summary statistics of inventory, performance, and utilization) ***L3*** (i.e., sufficient detail to conduct elementary methods of maintenance) ***L2*** (i.e., sufficient for comprehensive analysis) ***L1*** (i.e., most comprehensive)	FHWA report: "Asset Management Data Collection for Supporting Decision Processes" (*32*)
Data collection features[a]	Location (i.e., indexed location of the asset); dimension; material type and properties: asset condition	na	na
Data collection/construction phase[a]	Design; construction; project close-out (as-built); maintenance	na	na
Data collection techniques[a]	Manual collection; automated collection; remote collection	***Manual collection*** can use handheld computers, Global Positioning System (GPS) units, or pen-and-paper records. ***Automated collection*** involves a vehicle equipped with technologies to identify and document transportation assets. ***Remote collection*** uses photo logs, video logs, and satellite images.	FHWA report: "Roadside Safety Hardware Identification Methods" (*16*)
Data collection frequency	Monthly; quarterly; semi-annually; yearly; more than a year	na	na

Note: PDF = portable document format; 2D = two-dimensional; 3D = three-dimensional; FHWA = Federal Highway Administration; na = not applicable.
[a]Respondents were allowed to select more than one variable level.

implement an AMS for managing the asset data but is considering implementing one in the future).

2. Nature of the implemented AMS, to capture to what extent the asset data is structured and stored (scanned copies, spreadsheets, smart PDFs, database, or GIS database; public software; and private software).
3. Modeling capabilities of the adopted AMS, to capture if the adopted AMS has any data analysis capabilities (C1, C2, C3, C4, and C5).
4. Asset data flow paths between different divisions of the state DOTs to capture the asset data flow and how data is shared between different divisions (design to construction, construction to road inventory, construction to operation, and road inventory to and from operations).
5. Desired maintenance management approach to capture the optimized asset maintenance management approach and the needed data for efficient asset performance (condition-driven plus [optimal], condition-driven, cyclic-driven plus; cycle-driven; reactive; and minimal maintenance).
6. Consider a risk-based asset management approach to capture if the agency considers a risk

management approach for the considered asset to prioritize ancillary assets (a risk-based approach is considered for managing the asset, and a risk-based approach is not considered for managing the asset).

The different ancillary asset data management practices related to variables and variable levels are defined and summarized in Table 2.

Case Study

The case study was captured through an interview with subject matter experts, where an interview provides an opportunity for in-depth exploration of the topic, particularly when it is complex and lacks clarity (*34*). Furthermore, interviews are distinguished by the high level of interaction between the researcher and the participants (*34*). The interview was conducted with two subject matter experts. A principal engineer responsible for managing architecture, engineering, and construction (AEC) applications, developing processes for capital project delivery, incorporating technology and intelligence, and managing asset inventories with 37 years of experience, and a transportation engineer/BIM specialist with the computer-aided design (CAD) administration team with 27 years of experience. The interview was approved by the IRB at the University of Kentucky (IRB # 69802), and the consent of the interviewees to record the virtual meeting was obtained. The interviews were conducted online for a total of 240 min.

Table 2. Ancillary Asset Data Management-Related Variables and Variable Levels

Variable	Variable level	Variable level's definition	Reference if applicable
Asset data is managed using an asset management system (AMS)	The agency adopted an AMS for managing the asset data; the agency did not implement an AMS for managing the asset data; the agency did not implement an AMS for managing the asset data but is considering implementing one in the future	na	na
Nature of the implemented AMS[a]	Scanned copies, spreadsheets, smart PDFs, database, and/or GIS database; public software; private software	na	na
Modeling capabilities of the adopted AMS	C1 C2 C3 C4 C5	***C1*:** the adopted AMS has limited data modeling and analysis capabilities ***C2*:** the adopted AMS has some basic capabilities to model future conditions ***C3*:** the adopted AMS provides a variety of simple analysis scenarios and reporting methods ***C4*:** the adopted AMS uses agency-specific models to predict future conditions ***C5*:** the adopted AMS provides the most robust analysis and reporting	FHWA report: "Asset Management Data Collection for Supporting Decision Processes" (31)
Asset data flow paths between different divisions of the state DOT[a]	Design to construction; construction to road inventory; construction to operation; road inventory to and from operations	na	na

(continued)

Table 2. (continued)

Variable	Variable level	Variable level's definition	Reference if applicable
Desired maintenance management approach	Condition-driven plus (optimal); condition-driven; cyclic-driven plus; cycle-driven; reactive; minimal maintenance	***Condition-driven plus (optimal)*:** the condition of the asset is routinely monitored and modeled. Actions are taken proactively and reactively to optimize the asset lifecycle through minimum lifecycle cost, maximum benefit, maximum lifecycle length, or similar approaches; ***Condition-driven*:** the asset's condition is routinely monitored, and actions are taken to manage the long-term performance of the asset or the asset's impact on system performance. ***Cyclic-driven plus*:** the asset is inspected and maintained on a cyclical basis, where the inspection and maintenance activities are performed simultaneously or in concert, and condition data is collected and maintained for analysis that could lead to additional condition-based decisions. Condition data required. ***Cycle-driven*:** the asset is maintained on a cyclical basis. Condition data may be collected on these assets to meet other business needs, but the inspection cycle is managed separately from the maintenance cycle, and condition data is not required. ***Reactive*:** an inventory is maintained, but there is no regular condition data collection and no maintenance performed to slow or address damage or deterioration until an asset is reported as having an unacceptable defect. Annual work is planned at the aggregate level without concern for the specific locations of potential future defects. ***Minimal maintenance*:** no inventory or condition data is collected or maintained. Maintenance is performed when assets are identified as having an unacceptable defect.	Minnesota DOT (MnDOT) Asset Management Strategic Implementation Plan (*33*)
Consider a risk-based asset management approach	A risk-based approach is considered for managing the asset; A risk-based approach is not considered for managing the asset	A ***risk-based approach*** is considered when the impact of the failure is much greater than the cost of the asset failure and is significant to safety or operations	na

[a]Respondents were allowed to select more than one variable level.

Note: PDF = portable document format; GIS = geographic information system; FHWA = Federal Highway Administration; DOT = department of transportation; na = not applicable.

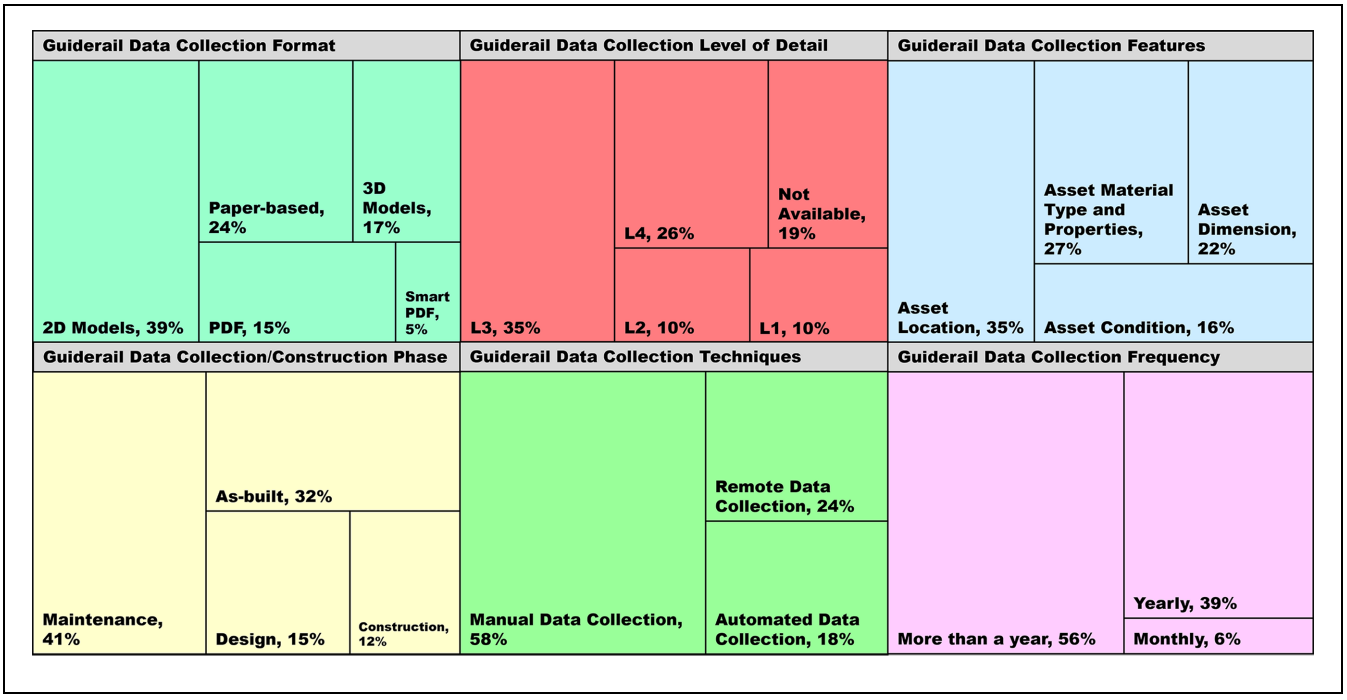


Figure 3. Current state of practice of state departments of transportation in collecting guiderail asset data.
Note: 2D = two-dimensional; 3D = three-dimensional; PDF = portable document format.

Results

Current Practices of State DOTs in Collecting and Digitizing Guiderail Asset Data

Out of the 45 collected responses, 38 respondents (i.e., 84% of the responses) indicated that they collect data for the guiderail ancillary asset class. This further emphasizes the critical role played by guiderails to improve roadway safety. The collected data were aggregated and analyzed, and the percentage of responses for each variable level was calculated. Figure 3 summarizes the current state of practice of state DOTs in collecting guiderail asset data for the six ancillary asset data collection variables and variable levels investigated, and the key findings are discussed in the following.

Data Collection Format. The current state of practice of state DOTs on the collected data format for guiderails indicates that state DOTs mainly adopt 2D models (39% of responses), followed by paper-based ones (24% of responses). In comparison, smart PDF is the least adopted format for guiderail data collection (5% of responses).

Data Collection Level of Detail. The current state of practice of state DOTs on the collected level of detail for guiderails indicates that state DOTs mainly adopt a level of detail L3 for guiderail data collection (i.e., sufficient detail to conduct elementary methods of maintenance) (36% of responses), followed by a level of detail L4 (i.e., basic details with summary statistics of inventory, performance, and utilization) (26% of responses). However, the results indicate that state DOTs do not collect guiderail asset data with higher comprehensive levels of detail, that is, L2 and L1, where the percentages for adopting these two levels pertain to less than 10% of the responses.

Data Collection Features. The current state of practice of state DOTs on the collected data features for guiderails, indicates that the location (i.e., the indexed location of the asset) is mainly collected (34% of response), followed by the material type and properties (27% of responses), then followed by the asset dimension (22% of responses). In comparison, the asset condition is the least-collected data feature for guiderails (17% of responses).

Data Collection/Construction Phase. The current state of practice of state DOTs on the data collection/construction phase indicates that state DOTs mainly collect guiderail asset data during the maintenance phase (41% of responses), followed by the project close-out or as-built data (32% of responses). In contrast, during the construction and design phase of the guiderail construction process, state DOTs usually do not collect data. The



Figure 4. Current state of practice of state departments of transportation (DOTs) in managing guiderail asset data.
Note: GIS = geographic information system.

percentages for the state of practice for guiderail data collection during these construction project phases are relatively low, 15% and 12% of responses, respectively.

Data Collection Technique. The current state of practice of state DOTs on the adopted method for data collection indicates that state DOTs mainly use manual data collection methods to collect guiderail asset data (58% of the responses), followed by remote data collection techniques (24% of responses), while automated data collection is the least adopted with 18% of responses.

Data Collection Frequency. The state of practice of state DOTs on the guiderail data collection frequency indicates that state DOTs mainly collect guiderail asset data over time intervals longer than a year (56% of the responses), followed by a yearly time interval (39% of responses). However, the results indicate that state DOTs do not collect guiderail asset data semi-annually or quarterly, and barely collect guiderail data monthly (6% of responses).

Current Practices of State DOTs in Digitalizing the Management of Guiderail Asset Data

The complete digital transition of an asset is integrating its data into a well-established electronic management system. It is essential to emphasize that a true digital transition does not merely involve replacing pen and paper with a smart screen; instead, it entails adopting an information management system with automated or semi-automated data acquisition capabilities (*9*). Figure 4 summarizes the state of practice of state DOTs in managing guiderail asset data for the six data management variables and variable levels investigated. The key findings are discussed in the following.

Adoption of Asset Management System for Managing Guiderail Data. Based on the 30 surveyed state DOTs, 14 state DOTs reported using an AMS to manage ancillary asset data. The survey also asked which asset classes were managed using an AMS, and 10 of the 14 DOTs (71%) reported using an AMS to manage guiderail data.

Nature of the Adopted AMS for Managing Guiderail Data. State DOTs commonly use a database including scanned copies, spreadsheets, smart PDFs, and/or a GIS database to manage guiderail asset data. This was reported by 86% of the respondents. Meanwhile, 29% of the respondents stated that they use private software (i.e., commercial off-the-shelf [COTS]) to manage their guiderail asset data. None of the state DOTs reported using publicly available software for this purpose.

Modeling Capabilities of the Adopted AMS for Managing Guiderail Asset Data. Most state DOTs use an AMS with limited data modeling and analysis capabilities, specifically at level C1 (83% of responses). However, the remaining 17% of responses indicated that the adopted AMS for managing guiderail assets offers various simple analysis scenarios and reporting methods with a modeling capability at level C3.

Guiderail Asset Data Flow Paths Between the Different Divisions Within a State DOT. Based on the responses, it was found that the majority (67%) of guiderail asset data flows between roadway inventory teams and operation teams. However, this data has minimal flow between design and construction teams (17%), construction and road inventory teams (17%), and construction and operation teams (17%).

Desired Maintenance Management Approach for Managing Guiderails. According to the respondents, the desired management approach for managing guiderail data is the "cyclic-driven plus" approach, which was chosen by 50% of the respondents. This involves inspecting and maintaining the guiderail on a cyclical basis, with the inspection and maintenance activities carried out simultaneously or in concert. Data on the guiderail's condition is collected and maintained for analysis, which could lead to additional condition-based decisions. It is emphasized that guiderail condition data is necessary for this approach. The remaining 50% of responses were distributed over the other investigated desired maintenance approaches.

Considering a Risk-Based Approach to Manage Guiderails. Implementing a risk-based approach for managing guiderail assets is desirable, as 60% of respondents have shown an inclination toward it.

Guiderail Data Architecture, Data Governance, and Stewardship Frameworks

After capturing and documenting the current practices of state DOTs in collecting and managing guiderail data, it is critical to understand the guiderail asset data architecture, asset data governance, and asset data stewardship frameworks. As such, an interview was conducted with subject matter experts from CTDOT.

Guiderail Data Architecture. Understanding data architecture is fundamental for an enterprise since it creates and maintains organizational knowledge about data and its lifecycle. This understanding supports enterprise data management and leveraging the asset data's value to optimize asset performance and decrease the risk of asset failure (*35*). The first section of the interview focused on documenting the guiderail asset data lifecycle and achieving a better understanding of the guiderail asset data architecture. The deployed asset data lifecycle framework was adopted from Ammar et al. (*36*), where the life of a transportation asset commences with its construction. It undergoes several rehabilitation and renewal activities, which would occur as maintenance work orders before eventually deteriorating over time or because of natural forces. However, data related to the asset extends before its construction phase, originating from the conceptual asset design and planning stages and continuing through to the asset construction phase to reflect its as-built condition. Moreover, this asset data continues to expand and support various operations and maintenance activities, including rehabilitation or replacement, until the asset becomes obsolete. At this point, the associated data is archived or disposed of. As a result, most asset data accumulates during the operation and maintenance phase of the asset's lifecycle, where it is widely recognized that operation and maintenance represent the largest expenditure and value within the transportation system (*36*).

The interviewees were asked to apply the adopted transportation asset data lifecycle ecosystem (TADLE), as presented in Figure 5 (*36*), for guiderails. The TADLE is initiated with the first stage of the guiderail asset data lifecycle, the *asset design data model*. During the asset design data model, the guiderail data will mainly constitute a CAD model that will include design data features such as the status or work type. For instance, if the asset is newly constructed or rehabilitated, the pay or the bid item and data features, such as the item description and the asset's exact geo-referenced location, are necessary. Such data will support the fabrication of guiderails by using the design data features stored within the CAD model, such as the asset dimensions, material, description, how the asset will be paid for, and the asset location, to enable better asset location in the field. The next phase of the guiderail asset data lifecycle is the *asset production data*, where the asset data will pass through a temporary pending inventory status before the asset is given an identification number within the implemented GIS inventory databases. After this temporary stage, the

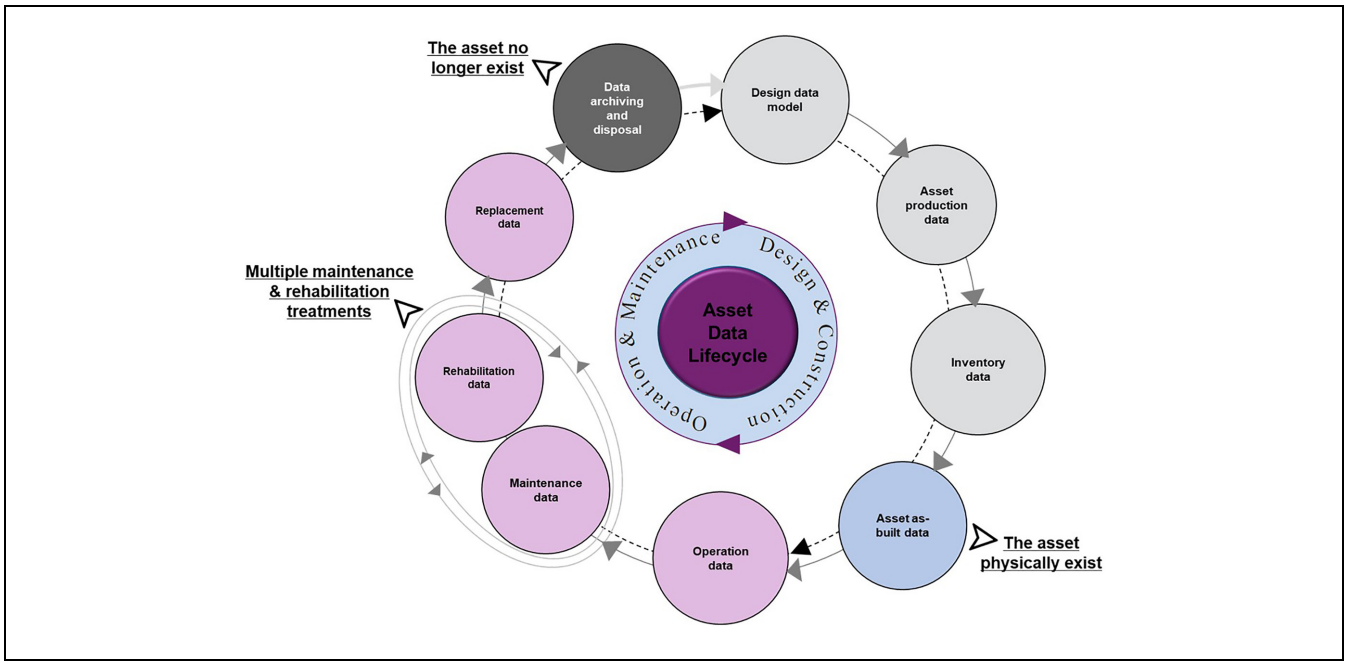


Figure 5. Transportation asset data lifecycle ecosystem.
Source: Adopted from Ammar et al. (*36*).

data related to the designed guiderail will be stored and referred to using a unique identification number, associated work types, and project information, all saved within the GIS inventory databases. The guiderail asset data is stored and can be easily extracted using the unique identification number or the asset geolocation. For instance, if this is a new construction, the guiderail can be installed, and the asset will physically exist.

After the guiderail asset is placed into service, the guiderail asset data will be referred to as the *as-built guiderail asset data*. The as-built asset data should capture information related to validating the asset's exact installation geolocation and any updates that the asset went through during the asset installation and would require adjustment of the previously referred to "design data model," where any changes in the asset condition should be documented in the now "as-built data model." If the change is because of a change in the contract, the updates are captured using a change order; however, if the changes were conducted in the field, the updated information should be captured and documented using a daily work report (DWR).

Following the asset's as-built data, the guiderail asset lifecycle will be initiated, and the guiderail asset will be operated and maintained through multiple maintenance and rehabilitation treatments. The *guiderail operation data* will include information related to the scheduled maintenance activities, roadway inspection, and asset condition data. For instance, it is important to capture if the change in the guiderail condition is caused by crashes and the frequency of occurrences, since this might reflect a highway safety problem that requires design intervention. Moreover, the guiderail asset maintenance and rehabilitation data will capture work order information and cost information, such as the cost of the workforce and tools or equipment used to maintain the guiderail. This type of information is usually saved in the asset maintenance management system (MMS).

Conversely, if the installed guiderail becomes obsolete and requires replacement, the guiderail replacement could take two routes: if the guiderail asset is replaced through a capital program (i.e., designed and paid through a capital program), the asset data lifecycle is reinitiated with the design data model stage. However, suppose the guiderail asset is replaced through a maintenance work order. In that case, maintenance workers will replace the obsolete guiderail, and the new guiderail will be given the same old identification number. Thus, even if the guiderail asset no longer physically exists, the corresponding data will be archived, and the data of the obsolete guiderail will be decommissioned from the inventory. Asset history information is recorded in the inventory with information related to the prior asset, the initial date of installation, service life, removal condition, asset materials, and cost of preservation throughout the guiderail lifecycle. The adopted TADLE framework for guiderails is a comprehensive approach to present and summarize the information required at each phase of the guiderail asset data lifecycle, which should be readily available for the following phase and would allow

for efficient operation and maintenance of the guiderail asset throughout the asset lifecycle.

Guiderail Data Governance and Data Stewardship Frameworks

Data governance outlines the responsibilities and accountability pathways concerning asset data management to fulfill its business objectives and ensure adherence to relevant legislation, regulations, and business standards (*37*). Conversely, data stewardship refers to the designated and entrusted lines of accountability for business responsibilities, distinct from IT responsibilities. The primary goal of data stewardship is to ensure the effective control and utilization of data and information assets (*37*). As such, the second section of the interview focused on comprehending the guiderail asset data governance and data stewardship frameworks.

To ensure the fulfillment of the CTDOT mission of providing a safe, efficient, and cost-effective transportation system that meets the mobility needs of its users, CTDOT's objective is to develop and adopt asset data governance and data stewardship frameworks that would enable the complete population of intelligence by including all the necessary and applicable data that support informative insights in the asset design model, that is, at the first stage of the asset data lifecycle and critical to flow to the following stages, for instance the asset geolocation. Also, CTDOT is establishing an asset data governance framework based on a well-structured foundation for the "digital as-built" data of the asset. In the context of the overarching data governance framework, they developed a set of policy instruments, including a digital project development manual (DPD), to support standardizing the asset design model. Within the developed DPD manual, CTDOT has a dedicated section called electronic engineering data (EED). This section outlines the specifications for consultant-led and in-house designed projects, providing guidance on the content to be included in the design model with respect to 2D and 3D elements. EED will be continuously updated to align with the latest version of newly adopted software, such as OpenRoads, and utilize data dictionaries developed by other state DOTs, such as those developed by the Florida DOT and UDOT. The aim is to enhance the visual presentation by adopting a spreadsheet-like format and incorporating distinct levels of detail. For instance, adopting a new software highlighted the importance of making the data requirements in CAD more user-friendly and understandable. It is critical to ensure that users clearly understand how the data should appear in the design model and the significance of certain data components. The interviewees also highlighted that even though relevant data governance guidelines are in place, continuous improvements are being made to align the current practices with the desired outcomes.

To implement data stewardship frameworks, CTDOT has a chief data officer and designated asset stewards responsible for managing specific assets. The chief data officer is vital in providing support in various capacities. However, there is a need to clarify the roles and responsibilities of these asset stewards for future improvement. Their primary task is to ensure that the data related to asset conditions is regularly updated to support effective asset management. This emphasizes the importance of data management and assigning enough responsibility to the asset stewards, where maintaining an accurate and up-to-date inventory is essential. In addition to data storage and acquisition, other important factors must be considered when managing resources. For instance, since decision-making is critical, assessing new data sets to capture asset conditions, such as aerial imagery, might be needed. This responsibility primarily falls on the data steward in charge.

Furthermore, CTDOT is currently setting targets for 2024 and improving its MMS to include more asset information and a wide variety of ancillary assets, namely guiderails. In addition, CTDOT is piloting a project to implement BIM with guiderails and exploring how to manage the underlying data by incorporating attribution into early design models. This allows inspectors to avoid dealing with attributes unless changes or specific issues arise. CTDOT is developing processes that enable them to write back information to their assets by creating a common data environment (CDE) that integrates project management information, project information models, and asset information models. The developed CDE supports the integration with a project estimation software (e.g., AASHTOware) and an internally developed enterprise GIS application (ATLAS 2.0 ESRI WebMap App). This approach enables CTDOT to continuously update its processes as they advance in managing different assets. CTDOT is currently investigating the implementation of the developed approach to managing guiderails and how it affects multiple stakeholders with different responsibilities.

Discussion

Implementing Digital Twins to Support the Management of Guiderail Assets

Digital Twins represents an innovative solution to manage assets throughout their lifecycle by collecting and integrating asset data to improve the design, construction, operation, and maintenance of infrastructure assets, resulting in the sustainable development of infrastructure systems. Data is the core of the definition of Digital Twins for transportation assets, where the asset data

plays a crucial role in connecting the physical asset to its digital representation. However, emphasizing the importance of data does not imply that Digital Twins should be constructed using an overwhelming number of databases that are challenging to understand and process. Moreover, implementing Digital Twins for asset management does not necessarily require that all assets be represented in 3D models; in contrast, Digital Twins should be built on databases containing high-quality information that is easily accessible, extractable, updatable, and integrable (*9*). As such, state DOTs need to invest in improving asset data quality, prioritize assets that should be represented in 3D models, and identify assets that can be represented using comprehensive 2D models with the necessary data attributes to manage assets. Ammar et al. (*38*) referred to high-quality data as informative data that enhances analytical asset models, such as asset inventory data, asset geolocation, and updated asset condition data, among other data attributes. These data elements are essential for extracting informative insights for creating predictive and preventive maintenance strategies and providing valuable insights for decision-making.

The findings from the conducted survey showed that 50% of the respondents indicated that the desired guiderail asset management approach is "cyclic-driven plus," where it is important to collect condition data and regularly inspect and maintain guiderails. Guiderail condition data should also be associated with the scheduled maintenance activities, roadway inspection data, work order information, and cost information needed to maintain the guiderail. State DOTs need to improve the level of detail of the collected data to include comprehensive asset data (i.e., level of detail L2). This can be enabled by collecting guiderail data using automated and remote technologies such as LiDAR and aerial imagery. It can also be achieved by adding necessary attributes to the asset design model and collecting the as-built asset data. Moreover, findings indicated a risk-based management approach could be considered to manage guiderails, thus emphasizing the importance of the asset in enhancing roadway safety. As such, the guiderail is a suitable candidate for implementing Digital Twins. Ammar et al. (*9*) reported the same findings.

A key step of creating a Digital Twin for managing guiderail assets is understanding and identifying the data types and attributes needed. This research indicates that guiderail data sets should include the asset design model, which can be a comprehensive 2D model enriched with data attributes, including design data features, pay or bid item, asset exact geolocation, asset dimension, material, and description. Using extraction, transformation, and loading (ETL) processes, relevant project information from the asset design model, which could be exported from BIM, is integrated with GIS inventory databases, the asset's unique identification number, and the associated work types.

Establishing asset data management maturity is critical to optimize guiderail management and successfully implement Digital Twins. This will enable the inclusion of informative data in the guiderail asset design model and allow for better data architecture and governance throughout the asset data lifecycle. Asset data maturity and Digital Twins implementation can be achieved by creating a CDE that provides a single version of asset data and enables all parties to communicate and collaborate securely and integrate data from multiple resources, such as BIM and GISs. ISO19650 3.3.15 defines CDE as an "agreed source of information for any given project or asset, for collecting, managing and disseminating each information container through a managed process" (*39*). As such, CDE is necessary for creating a Digital Twin and is an essential outcome for asset data flow and integration throughout the asset lifecycle, where data extracted from different management systems, such as project management, asset information, cost, and project estimation, must be included. Thus, collecting, storing, and managing guiderail asset data within an established CDE would improve the asset lifecycle operation and maintenance. The established CDE should be able to integrate data sets that can be easily extracted, transferred, and loaded back to the developed data environment. Furthermore, it is essential to have established asset management information systems (AMISs) where informative insights can be extracted, thus enabling state DOTs to reap the benefits of implementing Digital Twins and conduct predictive and preventive maintenance, in addition to developing asset deterioration analysis to enable cross-asset data-driven decisions supporting multiple objectives.

Conclusions

Digital Twins have emerged as a transformative technology with immense potential for enhancing asset data management across various industries, particularly transportation. The proper functioning and maintenance of safety hardware assets, such as guiderails, are critical for ensuring the safety of road users. However, traditional asset data management practices often lack real-time, comprehensive insights into these crucial assets. As such, this study investigated the current practices of state DOTs to collect and manage ancillary asset data with a primary focus on guiderails by developing and distributing a web-based survey. A case study was also conducted to provide a detailed understanding of the guiderail data architecture, governance, and stewardship. This understanding will enable the development of information-rich analytical models for guiderails using Digital Twins and

identifying relevant asset data throughout the asset lifecycle.

The responses from the survey showed that guiderail data could be presented using 2D models. However, high-quality asset data, including asset inventory, geolocation, updated condition data, and other attributes, is crucial in enhancing analytical asset models. It is important to include attributes in the asset design model in addition to collecting the as-built asset data (i.e., during the asset construction close-out). Also, the survey indicated that guiderails should be managed using an established AMS with modeling capabilities that can provide informative insights. The preferable guiderail asset maintenance management approach can be a cyclic-driven plus, where guiderails should be maintained cyclically, and the inspection and maintenance activities should be performed simultaneously. Condition data is collected and maintained for analysis that could lead to additional condition-based decisions.

Implementing Digital Twins for managing guiderails requires condition data and integrating automated and remote technologies such as LiDAR and aerial imagery for comprehensive data collection. Also, establishing asset data management maturity is essential to optimize guiderail management through Digital Twins. This involves creating a CDE and integrating data extracted from BIM and GISs. Data from various management systems, such as project management, asset information, cost, and estimation, must be included in the CDE. Moreover, adhering to ISO 19650 guidelines for managing information containers in a managed process facilitates efficient data collection, storage, and dissemination, ultimately improving the asset lifecycle's operation and maintenance. Successful implementation of Digital Twins in transportation asset management requires established asset management information systems capable of extracting informative insights. This enables the benefits of predictive and preventive maintenance, asset deterioration analysis, and data-driven decision-making with multiple objectives and contributes to the future vision of ITSs.

Future research will investigate the implementation of Digital Twins for other critical transportation assets. Furthermore, it will develop smart asset data management plans to guide state DOTs in navigating the journey of digitizing and digitalizing their asset management processes to successfully implement Digital Twins for transportation asset management.

Author Contributions

The authors confirm their contribution to the paper: study conception and design: A. Ammar, F. Maier, G. Dadi; data collection: A. Ammar; analysis and interpretation of results: A. Ammar, W. Pratt, E. Richard; draft manuscript preparation: A. Ammar. All authors reviewed the results and approved the final version of the manuscript.

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